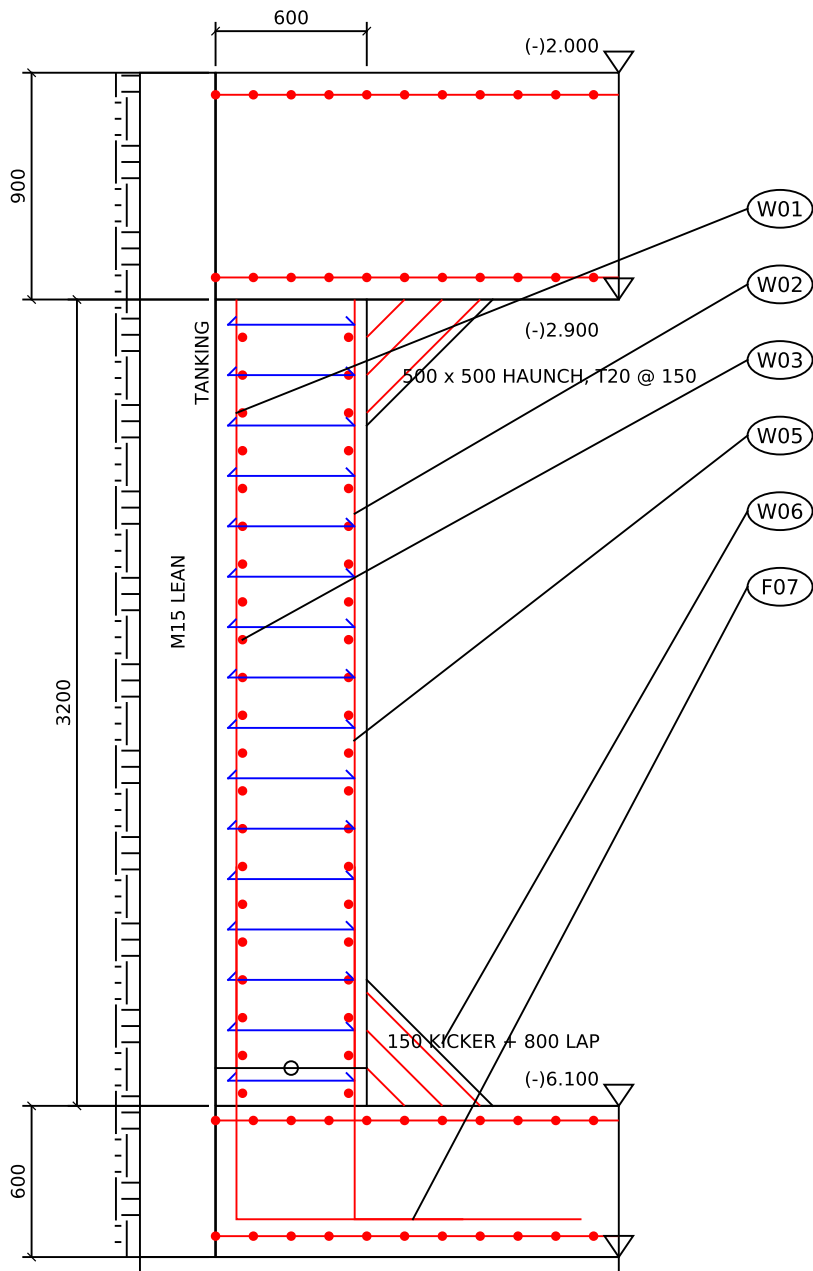
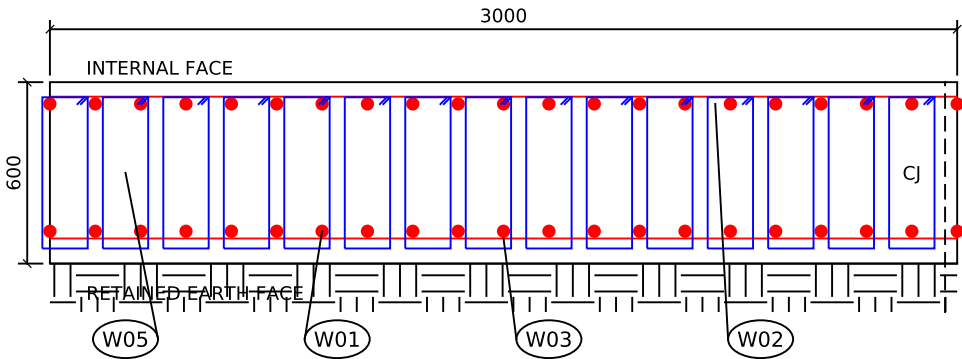




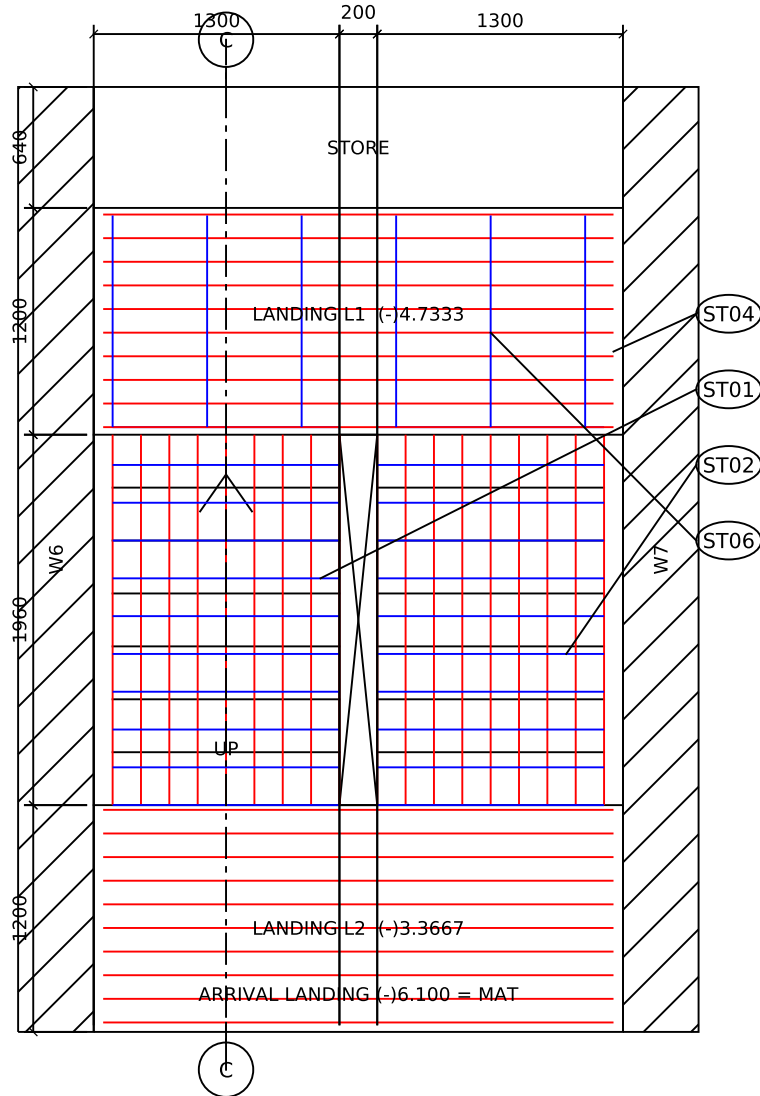
1 600 SHEAR WALL - VERTICAL SECTION
1 : 30 W1 / W2 / W3 / W4. T16 @ 150 EF EW, COVER 75 / 40



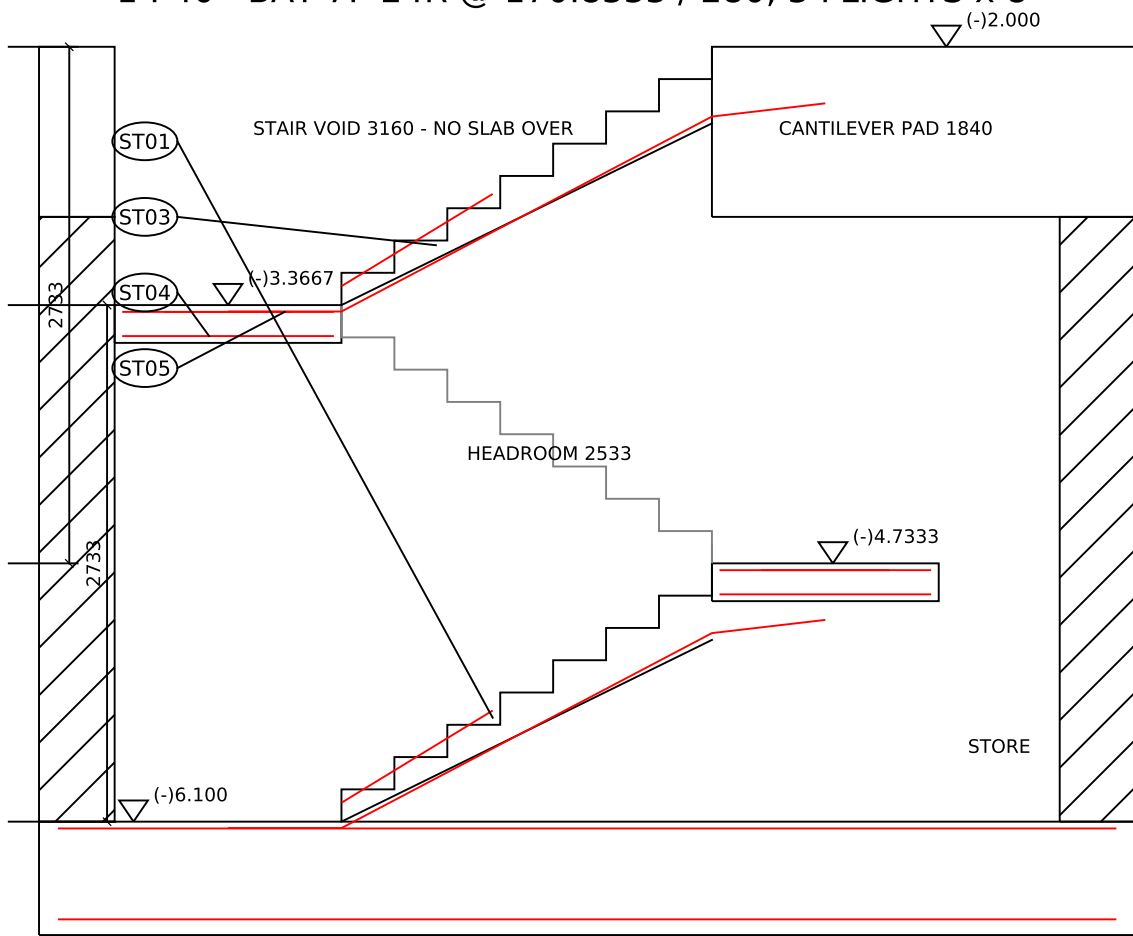
2 600 SHEAR WALL - PART PLAN
1 : 25 HORIZONTAL SECTION AT (-)4.500

WALL AND STAIR - CONSTRUCTION REQUIREMENTS

- WALL STARTERS F07: T16 @ 150 EACH FACE WITH A 900 mm HORIZONTAL LEG INTO THE MAT, lapped 800 (50 phi) above a 150 kicker. The starter, not the wall bar, is what makes the mat / wall joint work.
- HAUNCHES 500 x 500 AT EVERY WALL / ROOF AND WALL / MAT JUNCTION, with diagonal T20 @ 150 (mark W06).
- TWO WATERSTOPS AND A WELDED Cu / GALVANISED EMP STRAP AT EVERY CONSTRUCTION JOINT (approximately 6 m centres). Reinforcement is FULLY CONTINUOUS through the joint - it is not a movement joint.
- COVER: 75 mm to a face cast against the blinding or trimmed rock; 50 mm to a formed earth face; 40 mm internally; 30 mm in the stair shaft. The 5 mm reduction IS 456 Table 16 permits for M35 and above is DELIBERATELY NOT TAKEN.
- MAX BAR SPACING 150 IN BOTH CURTAINS - AN EMP REQUIREMENT, STRICTER THAN IS 456 Cl. 26.3.3. No bar may be displaced to clear a service; any clash is an RFI.
- THE MAIN STAIRCASE IS DESIGNED TO IS 456 WITH NORMAL PARTIAL FACTORS. It is inside the protective envelope and is NOT a blast element.
- STAIR FINISHES AND THE 1100 GUARDING TO THE VOID FREE EDGE (NBC 2016 Pt 4) ARE NOT SHOWN ON THIS SHEET - see the architectural package.



3 MAIN STAIRCASE - REINFORCEMENT PLAN
1 : 40 BAY 7. 24R @ 170.8333 / 280, 3 FLIGHTS x 8



4 MAIN STAIRCASE - SECTION C-C
1 : 40 CUT ON FLIGHT A AT X 15900, LOOKING EAST

MAIN STAIRCASE - THE GEOMETRY IS FROZEN

24 RISERS at 170.8333 | TREAD 280 | 3 FLIGHTS x 8 RISERS | TOTAL RISE 4100
FLIGHT WIDTH 1200 | WELL 200 | WAIST 200 | HEADROOM 2533
LANDING L1 (-)4.7333 L2 (-)3.3667 ARRIVAL (-)6.100 = THE MAT SURFACE
TOTAL RISE CLOSES EXACTLY: (-)6.100 TO (-)2.000 = 4100 = 24 x 170.8333
NO RISER, TREAD, LANDING LEVEL, FLIGHT WIDTH, WELL OR HEADROOM ON THIS SHEET
MAY BE ALTERED. ANY CHANGE IS A NEW INSTRUCTION, NOT A DETAILING DECISION.

WALL SCHEDULE

WALL	THK	LENGTH	COVER	d	MAIN BARS	LINKS
W1 SOUTH PERIMETER	600	22000	75/50/40	517	T16 @ 150 EF EW	T12 CLOSED @ 200
W2 NORTH PERIMETER	600	22000	75/50/40	517	T16 @ 150 EF EW	T12 CLOSED @ 200
W3 WEST END	600	6200	75/50/40	517	T16 @ 150 EF EW	T12 CLOSED @ 200
W4 EAST END	600	6200	75/50/40	517	T16 @ 150 EF EW	T12 CLOSED @ 200

ALL FOUR 600 WALLS: xu/d = 0.089, UTILISATION 68 %, BLAST GOVERNS 4.6 : 1

OTHER WALLS - SCHEDULED HERE, DETAILED ELSEWHERE, NOT DRAWN ON THIS SHEET

W5 BAY 5 / 6	200	5000	40	154	T12 @ 150 EF EW	- NOMINAL
W6 BAY 6 / 7 (M1)	400	5000	50/40	342	T20 @ 150 EF EW	T12 4L @ 200
W7 BAY 7 / 8 (M1)	400	5000	50/40	342	T20 @ 150 EF EW	T12 4L @ 200
W8 PARTITIONS x 4	110	5000	25	-	A252 MESH B/F	- NON-STRUCTURAL

BAR MARK SCHEDULE - 600 PERIMETER SHEAR WALLS

MARK	BAR	SPACING	LOCATION	CUT mm	No.
W01	T16	150	VERTICAL, EARTH FACE	3875	360
W02	T16	150	VERTICAL, INTERNAL FACE	3875	360
W03	T16	150	HORIZONTAL, EACH FACE	11350	176
W04	T16	150	HORIZONTAL, EACH FACE	7380	88
W05	T12	200 x 200	CLOSED LINKS, 200 x 200 GRID	1540	4320
W06	T20	150	HAUNCH DIAGONALS, WALL/ROOF + WALL/MAT	2307	720

MAIN STAIRCASE - ELEMENT SCHEDULE

ELEMENT	THK	d	MAIN	DISTRIB.	TOP STEEL
FLIGHTS 1, 2, 3 (WAIST)	200	164	T12 @ 150 (754)	T10 @ 200	T12 @ 150, 800 IN
LANDING L1 (-)4.7333	200	164	T12 @ 125 BTM (905)	T10 @ 200	T12 @ 125, 900 EACH END
LANDING L2 (-)3.3667	200	164	T12 @ 125 BTM (905)	T10 @ 200	T12 @ 125, 900 EACH END
ARRIVAL LANDING (-)6.100	-	-	= THE MAT SURFACE	-	NO SEPARATE SLAB

BAR MARK SCHEDULE - MAIN STAIRCASE

MARK	BAR	SPACING	LOCATION	CUT mm	No.
ST01	T12	150	MAIN BARS (x3 FLIGHTS)	3496	24
ST02	T10	200	DISTRIBUTION (x3 FLIGHTS)	1140	36
ST03	T12	150	TOP STEEL, 800 INTO THE FLIGHT	1280	48
ST04	T12	125	BOTTOM MAIN	3200	20
ST05	T12	125	TOP, 900 FROM EACH SUPPORT	1380	40
ST06	T10	200	DISTRIBUTION	1140	30

MAIN STAIRCASE - NBC 2016 PART 4 GEOMETRY CHECK

ITEM	PROVIDED	LIMIT	VERDICT
RISER	170.8333 mm	NOT MORE THAN 190	PASS
TREAD / GOING	280 mm	NOT LESS THAN 250	PASS
FLIGHT WIDTH	1200 mm	NOT LESS THAN 1000	PASS - STRETCHER CAPABLE
RISERS PER FLIGHT	8	12 PREFERRED MAX	PASS
HEADROOM	2533 mm	NOT LESS THAN 2200	PASS
TOTAL RISE	4100 = 24 x 170.8333	(-)6.100 TO (-)2.000	CLOSES EXACTLY

UNDERGROUND CBRN-HARDENED
BLAST-RESISTANT PROTECTIVE
STRUCTURE + SENTRY POST
PUNE, MAHARASHTRA

NOTES

- ALL DIMENSIONS ARE IN MM. ALL LEVELS ARE IN METRES RELATIVE TO FINISHED SITE GRADE 0.000, NEGATIVE DOWNWARDS.
- WRITTEN DIM TO BE FOLLOWED ONLY. DRG NOT TO BE SCALED.
- THIS DRG IS TO BE READ IN CONJUNCTION WITH ALL RELEVANT ARCH AND STR DRGS.
- CONCRETE M35, w/c NOT MORE THAN 0.45. REINFORCEMENT Fe500D TO IS 1786:2008. Ld 40 PHI, LAPS 50 PHI, ALL STAGGERED SO THAT NOT MORE THAN 50 % ARE SPLICED AT ONE SECTION.
- THE 600 WALL IS NOT STRENGTH-GOVERNED (68 %). 600 IS SET BY COVER, CONGESTION, IS 3370 CRACK CONTROL, THE EMP DOUBLE CURTAIN AT 150 AND THE 1168 kN/m AXIAL PATH.
- THE MAIN STAIRCASE IS INSIDE THE PROTECTIVE ENVELOPE AND IS NOT A BLAST ELEMENT. IT IS DESIGNED TO IS 456 WITH NORMAL PARTIAL FACTORS, NOT WITH IS 4991 DYNAMIC STRENGTHS.
- IS 13920:2016 HAS BEEN CHECKED FOR THE BOX: Cl. 10.4 BOUNDARY ELEMENTS ARE NOT TRIGGERED (WALL SHEAR 0.063 N/mm²) AND NO DUCTILE-DETAILING CLAUSE GOVERNS ANY BAR ON THIS SHEET. CONTRACTOR TO CHECK AND VERIFY THE DRG BEFORE EXECUTION.
-

STRUCTURAL
REINFORCEMENT
DETAILING -
600 SHEAR WALL
& MAIN STAIRCASE

CBRN HARDENED
UG OPS ROOM

SHEET 07

Project number 01

Date 16 SEP 2026

Drawn by SYN 01

Checked by DR IR CHAUDHARI

STR007

Scale As indicated